



Cheat Sheet **Seaborn**

**Data visualization
made easy**



**Seaborn is a Python library
for creating informative
and attractive statistical
graphics built on top of
Matplotlib.**



Basic Plots

Create a line plot

```
sns.lineplot(data=df, x='x', y='y')
```

Create a scatter plot

```
sns.scatterplot(data=df, x='x', y='y')
```

Create a bar plot

```
sns.barplot(  
    data=df, x='category', y='value'  
)
```



Distribution Plots

Visualize a histogram

```
sns.histplot(data=df, x='value', bins=30)
```

Create a kernel density plot

```
sns.kdeplot(  
    data=df, x='value', shade=True  
)
```

Draw a boxplot

```
sns.boxplot(  
    data=df, x='category', y='value'  
)
```

Create a distribution plot

```
sns.displot(  
    data=df, x='value', kind='kde')
```



Categorical Plots

Create a count plot

```
sns.countplot(data=df, x='category')
```

Draw a violin plot

```
sns.violinplot(  
    data=df, x='category', y='value')
```

Create a swarm plot

```
sns.swarmplot(  
    data=df, x='category', y='value'  
)
```

Create a strip plot

```
sns.stripplot(  
    data=df, x='category', y='value',  
    jitter=True)
```



Heatmaps

Plot a correlation heatmap

```
sns.heatmap(  
    data=df.corr(), annot=True,  
    cmap='coolwarm'  
)
```

Create a pivot table heatmap

```
pivot_data = df.pivot_table(  
    index='x', columns='y', values='value')  
sns.heatmap(  
    data=pivot_data, cmap='viridis')
```

Masked heatmap for triangular matrix

```
mask = np.triu(np.ones_like(df.corr()))  
sns.heatmap(data=df.corr(), mask=mask)
```



Pairwise Relationships

Plot pairwise relationships

```
sns.pairplot(data=df)
```

Add hue to pairplot

```
sns.pairplot(data=df, hue='category')
```

Create a corner plot

```
sns.pairplot(data=df, corner=True)
```

Plot a joint distribution

```
sns.jointplot(  
    data=df, x='x', y='y', kind='hex'  
)
```



Regression Plots

Create a regression plot

```
sns.regplot(data=df, x='x', y='y')
```

Draw a residual plot

```
sns.residplot(data=df, x='x', y='y')
```

Create a lmpplot with facets

```
sns.lmpplot(  
    data=df, x='x', y='y', hue='category',  
    col='group'  
)
```



Faceted Plots

Create a FacetGrid

```
g = sns.FacetGrid(  
    data=df, col='category', row='group'  
)  
g.map(sns.scatterplot, 'x', 'y')
```

Create a categorical facet plot

```
sns.catplot(  
    data=df, x='x', y='y',  
    col='category', kind='box'  
)
```

Create a distribution facet plot

```
sns.displot(  
    data=df, x='value',  
    col='category', row='group', kind='kde'  
)
```



Advanced Customization

Set figure size

```
plt.figure(figsize=(10, 6))  
sns.lineplot(data=df, x='x', y='y')
```

Customize color palette

```
sns.set_palette('Set2')  
sns.lineplot(  
    data=df, x='x', y='y', palette='Set2'  
)
```

Control figure aesthetics

```
sns.set_context('talk')
```



Styling and Themes

Set Seaborn theme

```
sns.set_theme(style='whitegrid')
```

Add titles and labels

```
plt.title('Plot Title')  
plt.xlabel('X-axis Label')  
plt.ylabel('Y-axis Label')
```

Save a plot to file

```
plt.savefig(  
    'plot.png', dpi=300, bbox_inches='tight'  
)
```

Rotate x-axis labels

```
plt.xticks(rotation=45, ha='right')
```



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VISHNU VARDHAN